

Varad Deshmukh

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#) |

Portfolio: <https://varadportfolio-blue.vercel.app/>

SUMMARY

Information Technology undergraduate (B.Tech, PCCOE) with strong core fundamentals in Data Structures & Algorithms, Systems, and Full-Stack Engineering. Hands-on expertise in Java/Spring Boot backend architecture, Redis distributed caching, Next.js/Three.js WebGL rendering, and embedded systems (ESP32). Solved 500+ DSA problems across LeetCode and HackerRank. Eager to leverage software engineering and system optimization skills to build scalable enterprise solutions.

EDUCATION

Pimpri Chinchwad College of Engineering (PCCOE), Pune

B.Tech in Information Technology | **2023 – 2027**

CGPA: 8.17 / 10.0

Higher Secondary (12th HSC): 89%

Secondary (10th SSC): 83.4%

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, SQL, Embedded C/C++

Backend Development: Spring Boot, REST APIs, Microservices, JWT, OAuth 2.0, Redis, FastAPI

Frontend Development: Next.js, Three.js (WebGL), React.js, Tailwind CSS, HTML5/CSS3

Databases & Tools: PostgreSQL, SQL, Docker, Maven, Git, GitHub, AWS Fundamentals

Hardware / Embedded: ESP32, GPIO Programming, ADC, Bluetooth Interfacing, IoT Basics

Core Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks

PROJECTS

Distributed API Rate Limiter | *Java 21, Spring Boot, Redis, Lua, Docker, Maven*

#Engineered a distributed rate-limiting microservice utilizing a **Sliding Window Algorithm** to smooth out high-concurrency traffic bursts.

#Authored atomic **Redis Lua scripts** to eliminate race conditions across multi-instance application deployments.

#Implemented custom HTTP interception filters to enforce rate caps and return structured 429 Too Many Requests response headers.

Interactive 3D Developer Portfolio & Dashboard | *Next.js, Three.js, WebGL, React, Tailwind CSS*

#Built an interactive 3D web application featuring custom WebGL particle systems, custom shaders, and responsive UI micro-interactions.

#Optimized 3D rendering pipeline for lower-tier hardware by using **instanced meshes**, optimized particle buffers, and frame-rate throttling.

#Integrated real-time REST API fetchers to dynamically pull live competitive programming metrics and project analytics.

Advanced Finance Ecosystem | *Spring Boot, React.js, PostgreSQL, JWT, OAuth 2.0*

#Developed a full-stack financial tracking platform with dynamic multi-currency support and real-time spending analytics.

#Implemented secure authentication workflows using **Google OAuth 2.0** and stateless **JWT authorization**.

#Designed RESTful endpoints using Spring Data JPA for transaction management, budgeting modules, and interactive reporting dashboards.

CERTIFICATIONS & VIRTUAL EXPERIENCES

Oracle: AI Database Foundations Associate

JPMorgan Chase & Co.: Software Engineering Virtual Experience

Deloitte: Data Analytics Virtual Experience

Google: AI Essentials & AI Prompting Essentials

IBM Cognitive Class: SQL and Databases

ACHIEVEMENTS

Competitive Programming: Solved **500+ Data Structures & Algorithms problems** across LeetCode, HackerRank, and CodeChef.

Core Problem Solving: Strong practical grasp of Arrays, Dynamic Programming, Graphs, Trees, and System Optimization.